# MSc in Financial Mathematics, FM50 topic: A joint density for FX rates from observed cross-smiles

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This document describes one of the available topics for the MSc project in Financial Mathematics. It should be read in conjunction with the overall FM50 Financial Maths Project guide

## Background

For currencies like the Euro, Dollar and Pound, European options are very actively traded on all three cross-rates between them. A natural question to ask is: how can we construct a joint density for two of the cross-rates so as to be consistent with observed European call prices at multiple strikes (at a single maturity) on all three cross-rates.

Assume our home currency is dollars, and we let $X$ denote the price of 1 Euro in dollars at time $T$ in the future (this is known as the EUR/USD rate), and $Y$ denote the price of 1 pound in dollars at time $T$ (known as the GBP/USD rate). If we assume interest rates are zero for simplicity, then given a risk-neutral measure $\mathbb{Q}$, the risk-neutral price of a European call option with strike $K$ on $X$ is given by

$$\mathbb{E}((X-K)^+)$$

and the price of a European call option with strike $K$ on $Y$ is given by

$$\mathbb{E}((Y-K)^+)$$

where (here and henceforth) $\mathbb{E}(.)$ refers to expectation under the measure $\mathbb{Q}$. A European option on EUR/GBP (i.e. on the cross-rate $Z = X/Y$) pays $(Z-K)^+$ **pounds** at time $T$, or equivalently $Y(Z-K)^+$ dollars, so the initial price of such an option (in pounds) is

$$\frac{1}{Y_0}\mathbb{E}(Y(\frac{X}{Y}-K)^+) \quad = \quad \mathbb{E}^{\tilde{\mathbb{Q}}}((Z-K)^+) \tag{1}$$

where we have re-written the expectation on the left side here using a **new probability measure** $\tilde{\mathbb{Q}}(A) := \mathbb{E}(\frac{Y}{Y_0}1_A)$ associated with using GBP as the home currency (where $Y_0$ is the value of GBP/USD at time zero), which is a genuine probability measure as $\mathbb{E}(Y) = Y_0$ because $\mathbb{Q}$ is a risk-neutral measure and we are assuming zero interest rates.

Then from the **Breeden-Litzenberger formula**, if $X$ and $Y$ have a density under $\mathbb{Q}$, then these densities can be extracted from the call option prices as

$$\begin{aligned} \mu_X(K) &= \frac{d^2}{dK^2}\mathbb{E}((X-K)^+) \\ \mu_Y(K) &= \frac{d^2}{dK^2}\mathbb{E}((Y-K)^+) \end{aligned}$$

(proof uses Leibniz rule) and similarly the density of $Z$ **under the new measure** $\tilde{\mathbb{Q}}$ is

$$\mu_Z(K) \quad = \quad \frac{d^2}{dK^2}\mathbb{E}^{\tilde{\mathbb{Q}}}((Z-K)^+)\,.$$

Now suppose we are given three **target marginal densities** $\mu_X$, $\mu_Y$ and $\mu_Z$ on $(0,\infty)$ and let $\Pi(\mu_X,\mu_Y)$ denotes the space of **joint probability measures** on $\mathbb{R}^2_+ = (0,\infty)\times(0,\infty)$ with **marginal distributions** $\mu_X$ and $\mu_Y$ respectively. We wish to find a $\mu^* \in \Pi(\mu_X,\mu_Y)$ such that if $X, Y$ have joint distribution $\mu^*$, then $X/Y \sim \mu_Z$ under the probability measure $\tilde{\mathbb{Q}}(.)$ defined above. If we can do this, then from the three equations above for $\mu_X, \mu_Y$ and $\mu_Z$, $\mu^*$ will be consistent with call options at all strikes, on all three cross-rates. A solution to this problem (that we now describe) is a variant of the method originally introduced in [Guy20] for VIX options (see also [F26]).

The aforementioned target densities can be inferred from **European option prices** using e.g. the **Gatheral SVI parametrization** (see the formula in the first task of Part 2 below which is taken from [Gath04]), and other later SVI articles by Gatheral, Jacquier, Martini, Mingone et al. to **interpolate** between **market implied volatilities**, and the Breeden-Litzenberger formula described above. $\mu^*$ can then be used to price e.g. basket, quanto or best-of options whose terminal payoff depends on $X$ and $Y$ (see Task 3 below).

Let $\bar{\mu}(x,y)$ denote a **reference probability density**, typically chosen to be the product of the two densities i.e. $\bar{\mu}(x,y)=\mu_X(x)\mu_Y(y)$ so $\bar{\mu}$ automatically satisfies two of the three marginal constraints.

**Without loss of generality, we also assume that $\int_0^\infty x\mu_X(x)dx=\int_0^\infty y\mu_Y(y)dy=\int_0^\infty z\mu_Z(z)dz=1$ (since if this is not the case we can just divide $X$ by its initial forward price and similarly for $Y$, so the initial price of $Z$ is also 1 because $\mathbb{E}^{\tilde{\mathbb{Q}}}(\frac{X}{Y})=\mathbb{E}^{\mathbb{Q}}(X)=1$).**

As in [Guy20], we now guess a solution to the problem described above of the form

$$\mu^*(x,y) \quad = \quad e^{u(x)+v(y)+yw(\frac{x}{y})}\bar{\mu}(x,y) \tag{2}$$

for some functions $u$, $v$, $w$ defined on $(0,\infty)$. Integrating $\mu^*(x,y)$ in $y$ or $x$ we see that the marginal density constraints can be written as

$$\mu_X(x) \quad = \quad e^{u(x)}\int_0^\infty e^{v(y)+yw(\frac{x}{y})}\mu_X(x)\mu_Y(y)dy \tag{3}$$

$$\mu_Y(y) \quad = \quad e^{v(y)}\int_0^\infty e^{u(x)+yw(\frac{x}{y})}\bar{\mu}(x,y)dx \tag{4}$$

(note these marginal constraints automatically enforce that $\mathbb{E}(e^{u(X)+v(Y)+Yw(\frac{X}{Y})})=1$ so $\mu^*$ integrates to 1, because if the joint density was off by a constant factor then the marginals $\mu_X(x)$ and $\mu_Y(y)$ will be out by the same factor). To match the cross-rate option prices $\int_0^\infty (z-K)^+\mu_Z(z)dz$, using (1) and recalling that we have now set $Y_0=1$, we see that

$$\begin{aligned}\int_0^\infty (z-K)^+\mu_Z(z)dz \quad = \quad \mathbb{E}^{\mathbb{Q}}(Y(\frac{X}{Y}-K)^+) \quad &= \quad \int_0^\infty\int_0^\infty y(\frac{x}{y}-K)^+e^{u(x)+v(y)+yw(\frac{x}{y})}\bar{\mu}(x,y)dydx\\ &= \quad -\int_0^\infty\int_\infty^0 \frac{x}{z}(z-K)^+e^{u(x)+v(\frac{x}{z})+\frac{x}{z}w(z)}\frac{x}{z^2}\bar{\mu}(x,\frac{x}{z})dzdx\\ &= \quad \int_0^\infty\int_0^\infty \frac{x}{z}(z-K)^+e^{u(x)+v(\frac{x}{z})+\frac{x}{z}w(z)}\frac{x}{z^2}\bar{\mu}(x,\frac{x}{z})dzdx\\ &= \quad \int_K^\infty\int_0^\infty \frac{x}{z}(z-K)e^{u(x)+v(\frac{x}{z})+\frac{x}{z}w(z)}\frac{x}{z^2}\bar{\mu}(x,\frac{x}{z})dxdz\end{aligned}$$

where we have made the transformation $z=x/y$ in the second line (i.e. $y=x/z$, so $dy=-\frac{x}{z^2}dz$ in the inner integral, and we have switched the order of integration in the final line.

Then taking the derivative with respect to $K$ and applying the **Leibniz rule** to the right hand side and using that $(z-K)|z=K=0$ and $\frac{d}{dK}(z-K)=-1$, we get

$$-\int_K^\infty \mu_Z(z)dz \quad = \quad -\int_K^\infty\int_0^\infty \frac{x}{z}e^{u(x)+v(\frac{x}{z})+\frac{x}{z}w(z)}\frac{x}{z^2}\bar{\mu}(x,\frac{x}{z})dxdz$$

and taking the derivative wrt $K$ again, we see that

$$\mu_Z(K) \quad = \quad \int_0^\infty \frac{x}{K}e^{u(x)+v(\frac{x}{K})+\frac{x}{K}w(K)}\frac{x}{K^2}\bar{\mu}(x,\frac{x}{K})dx\,. \tag{5}$$

Setting $\bar{\mu}(x,y)=\mu_X(x)\mu_Y(y)$ as discussed above, (3) and (4) can be re-arranged as

$$u(x) \quad := \quad -\log\int_0^\infty e^{v(y)+yw(\frac{x}{y})}\mu_Y(y)dy \tag{6}$$

$$v(y) \quad := \quad -\log\int_0^\infty e^{u(x)+yw(\frac{x}{y})}\mu_X(x)dx \tag{7}$$

and setting $K=z$ in (5) and simplifying, we see that

$$\int_0^\infty e^{u(x)+v(\frac{x}{z})+\frac{x}{z}w(z)}\frac{x^2}{z^3}\bar{\mu}(x,\frac{x}{z})dx-\mu_Z(z) \quad = \quad 0 \tag{8}$$

**Three coupled integral equations**

but we cannot take the $\frac{x}{z}w(z)$ term outside the integral here since it also depends on $x$. As an easier warm-up problem, you can consider the Sinkhorn scheme with $yw(z)$ replaced with just $w(z)$ which will lead to an explicit 3rd Sinkhorn equation for $w$ as for $u$ and $v$.

The Sinkhorn iterative **fixed point** scheme to solve the set of coupled integral equations in (6), (7) and (8) is then given by

$$\begin{aligned}
u^{n+1}(x) &= -\log\int_0^\infty e^{v_n(y)+yw_n(\frac{x}{y})}\mu_Y(y)dy \\
v^{n+1}(y) &= -\log\int_0^\infty e^{u_{n+1}(x)+yw_n(\frac{x}{y})}\mu_X(x)dx \\
\mu_Z(z) &= \int_0^\infty e^{u_{n+1}(x)+v_{n+1}(\frac{x}{z})+\frac{x}{z}w_{n+1}(z)}\frac{x^2}{z^3}\bar{\mu}(x,\frac{x}{z})dx
\end{aligned}$$

**Picard** scheme from FM02. with $u^0 \equiv v^0 \equiv w^0 \equiv 0$, which typically converges very quickly. Note the final equation in (8) here is just a one-dimensional **root-finding** problem to solve for the unknown $w^{n+1}(z)$ for each $z$-value in $(0,\infty)$ and under suitable conditions one can verify that a unique root exists. The first two equations here are conceptually the same as the simple $x_{n+1} = g(x_n)$ fixed-point scheme for solving a general non-linear equation $x = g(x)$.

| EUR/USD | 1.0681 | 1.0791 | 1.0904 | 1.1014 | 1.1119 |
|---|---|---|---|---|---|
| Implied vol | 0.0554 | 0.053115 | 0.0516 | 0.051435 | 0.0523 |
| GBP/USD | 1.2456 | 1.2595 | 1.274 | 1.2883 | 1.3017 |
| Implied vol | 0.06055 | 0.058665 | 0.0573 | 0.057185 | 0.05765 |
| EUR/GBP | 0.84386 | 0.84969 | 0.85585 | 0.86234 | 0.86875 |
| Implied vol | 0.03809 | 0.03725 | 0.037225 | 0.03825 | 0.03986 |

**Divide all strikes by the associated forward price** Table 1: (mid) implied volatilities with 1-month maturity (with strikes going horizontally) on 16th Mar 2024. The forward prices here are 1.0903 for EUR/USD, 1.2738 for GBP/USD and 0.85590 for EUR/GBP (data obtained from Bloomberg).

## Part 1: Literature review

The most relevant citations to discuss in Part 1 are [F26] and [Guy20].

## Part 2: Numerical computations

- **Task 1**. Using a simple least squares minimization scheme, fit the **Gatheral**[Gath04] **SVI parametrization** for **implied volatility** $\hat{\sigma}(K)$: $\hat{\sigma}(K)^2T = a+b(\rho(x-m)+\sqrt{(x-m)^2+\sigma^2})$ (where $x = k = \log\frac{K}{F_0}$ is known as the **log-moneyness** and $F_0$ is the initial **forward price**) to the three implied volatility smiles in Table 1 for $T = \frac{1}{12}$ (i.e. 1 month maturity), and list your SVI parameters in the blank table below to 4 significant figures. Plot the three SVI smiles (with the original market values included) and their associated risk-neutral densities over an appropriate range, by applying the Breeden-Litzenberger formula to your SVI call price function, i.e. $\mu(K) = \frac{d^2}{dK^2}C^{\mathrm{BS}}(S_0,K,\hat{\sigma}(K),T)$ ($C^{\mathrm{BS}}(S,K,\sigma,T)$ here is the usual Black-Scholes call price function with $r = q = 0$). Numerically check that the densities you obtain are non-negative on the range you consider.

| SVI parameters | $a$ | $b$ | $\sigma$ | $\rho$ | $m$ |
|---|---|---|---|---|---|
| EUR/USD | | | | | |
| GBP/USD | | | | | |
| EUR/GBP | | | | | |

- **Task 2**. Implement the Sinkhorn algorithm (as described in the background section) to numerically compute a joint density $\mu^*$ for $X, Y$ of the form in (2) so as to be consistent with all 3 SVI smiles, using Gauss-Legendre quadrature to compute the single and double integrals in Eqs (6) to (8), and finally to check for consistency with the three smiles over a range of strikes comparable to those in the table below. We advise you to store `numpy` arrays for $u$ (of dimension $N \times 1$), and $v$ and $w$ (both of dimension $1 \times N$) where $N$ is the number of quadrature points (try e.g. 100, 200 or 400), and also fit a **cubic spline** to $w$, so you can then easily compute a matrix of values for $yw(x/y)$, and use `numpy.sum(...,axis=(0,1))` to sum an array quickly over two dimensions for the Gauss-Legendre quadrature, and to iterate the Sinkhorn scheme around 50 times, and you may need some truncation to avoid having the splines perform extrapolation rather than interpolation when evaluating $w(x/y)$. We recommend using `optimize.bisect(func, a,b)` to implement the bisection method for the root finding. You may find it helpful to cap the output of the `numpy.exp` function to just below the largest value that Python can handle ($\approx 709$) to avoid overflow while testing the code.

Try setting $a=-1000$, $b=100$ as the left and right endpoint for the bisection scheme

- **Task 3**. Plot $u$, $v$ and $w$ for a sensible range of values, and using your joint density $\mu^*$ for $(X,Y)$, price a **basket option** which pays $(\frac{1}{2}(X+Y)-1)^+$ dollars at $T$, a **quanto call option** that pays $(\frac{X}{Y}-1)^+$ dollars at $T$, a **best-of** (rainbow) option which pays $(\max(X,Y)-1)^+$ dollars at $T$ and a "quadratic" option which pays $(X-Y)^2$ dollars at $T$ (again using Gauss-Legendre quadrature), and add these prices into a table of the following form (you may assume interest rates are zero for simplicity so no need for discounting).

| $\mathbb{E}((\frac{1}{2}(X+Y)-1)^+)$ | $\mathbb{E}((\frac{X}{Y}-1)^+)$ | $\mathbb{E}((\max(X,Y)-1)^+)$ | $\mathbb{E}((X-Y)^2)$ |
|---|---|---|---|
| | | | |

Work with normalized strikes: $K/F_0$

You will likely need cubic splines in your code: `from scipy.interpolate import CubicSpline`

Gauss-Legendre quadrature approximates a 1d integral $\int_a^b f(x)dx$ with a sum of the form $\sum_{i=1}^n w_i f(x_i)$, or (similarly) a 2d integral $\int_a^b \int_c^d f(x,y)dxdy$ with $\sum_{i=1}^n \sum_{j=1}^m w_i w_j f(x_i, y_j)$ for optimally chosen weights and $x$ and $y$-values. Example us: use GL quadrature to integrate the standard Normal density:

```
def func(x): return (2*np.pi)**-0.5 * np.exp(-0.5*x*x) a=-5;b=5;
[u,w]=np.polynomial.legendre.leggauss(NumWeights)
%Transform to an integral on the standard interval $[-1,1]$
u1=.5*(b-a)*u+.5*(a+b);w1=(b-a)/2*w
print(np.sum(w1*func(u1)))
```

For the Sinkhorn task I suggest computing $z = x/y$, with $x$ as a column vector and $y$ as a row vector:

```
x_=x_.reshape((NumWeights,1))
y_=y_.reshape((1,NumWeights))
```

then `z=x/y` is a matrix. Plot $u, v, w$ over the range .9 to 1.1 as in [F26], even though in your code $z$ will span the larger interval of .9/1.1 to 1.1/.9.

```
%Build a cubicspline for the w function from $w$-values defined at the values in the array w_[0,:] , and
z_=z_.reshape((1,NumWeights))
wfun = CubicSpline(z_[0,:],w_[0,:])
w__=wfun(x_/y_)

%two-dimensional integration with Gaussian quadrature
tmp2=w_1*w_2*mubar(x_,y_)*np.exp(u_+v_+y_*w__)
c_=tmp2.sum(axis=(0,1))
```

Note the SVI parametrization has **constraints** on its parameters to prevent the SVI density from going negative (also known as **butterfly arbitrage**.

```
def SVI(x,params):
    a=params[0];b=params[1];rho=params[2];m=params[3];sigma=params[4]
    %Enforce that |rho|<=1
    rho=min(max(rho,-1.0),1.0)
    b=min(b,2/(1+np.abs(rho)))
    a=max(a,-b*sigma*np.sqrt(1-rho**2))
```

If using this approach, remember to output the actual values of rho, b and a which come out *after* the constraint modification lines in the code here not before.

## Part 3: Personal contribution

In this section you are expected to conduct further research on a topic/problem of your choosing, which is typically an extension or variant of Part 2. Part 3 should contain your own numerical results and code, and the topic should be discussed with your supervisor. Below we list some potentially interesting ideas to explore, but more ideas will be discussed in the first project meeting.

- One can also solve the problem in Task 2 as a constrained entropy minimization problem or an (unconstrained) concave maximization problem rather than using the Sinkhorn method (see `https://martinforde.github.io/FiniteOptionsCase.pdf` and [F26] for further details), and you can use e.g. Python/MOSEK for this.

- **Discretize** $\mu_X$ and $\mu_Y$ in Task 2 (or use the bid/ask implied vols given in the Table), and compute the minimal/maximal admissible price (and corresponding **sub/super hedge**) for an option with a general payoff $c(X, Y)$ as a **linear programming** problem using the `linprog` command in Python/MATLAB (see e.g. chapters 5-6 of FM14 on KEATS), or you can also include options on $Z = X/Y$ in the calibration set (this just requires populating a large matrix and `linprog` does the rest).

## References

[Dup94] Dupire, B., "Pricing with a smile", 1994, Risk, 7, 18-20.

[F26] Forde, M., "Solving the FX cross-smiles problem - rate of convergence for Sinkhorn marginals, and the finite-option case", to appear in *Stat. Prob. Lett.*

[Gath04] Gatheral, J., "A parsimonious arbitrage-free implied volatility parameterization with application to the valuation of volatility derivatives", presentation, Global Derivatives&Risk Management conference, Madrid, 2004

[Guy20] Guyon, J., "Dispersion-Constrained Martingale Schrödinger Bridges: Joint Entropic Calibration of Stochastic Volatility Models to S&P 500 and VIX Smiles", *Risk*, April 2020